

MEGHARAJ V S

Data Scientist | AI Engineer

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PROFESSIONAL SUMMARY

Data Scientist with 1.3+ years of production experience building LLM-powered AI agents, multi-path RAG pipelines, and cloud-native ETL systems across a network of 6,000+ EV chargers, serving 1,700+ daily active users and processing 3.5M+ transactions. Proven track record delivering anti-hallucination safeguards, role-based access governance, real-time streaming, and ML-driven pricing in safety-critical, high-throughput environments. Skilled in Python, LangChain, LangGraph, OpenAI API, AWS, and production-grade MLOps, with a strong record of designing and shipping reliable, scalable AI systems that drive measurable business impact.

TECHNICAL SKILLS

Category	Skills & Tools
Languages	Python, Java, C, C++, JavaScript
Databases	PostgreSQL, MongoDB, MySQL, Redis
Machine Learning	Scikit-learn, XGBoost, Random Forest, Facebook Prophet, TensorFlow, PyTorch, Keras, NLP, Computer Vision, Time-Series Forecasting, Clustering (KMeans), Feature Engineering
LLMs & AI Systems	LLM Tool Calling, RAG (BM25 + ChromaDB + Cross-Encoder Reranking), Prompt Engineering, LangChain, LangGraph, OpenAI API, AWS Bedrock (Claude), LLaMA/PEFT, Generative AI, Agentic Systems
Analytics & Visualization	Power BI, Matplotlib, Seaborn, Statistical Modeling, Time Series Analysis, Forecasting, PySpark, Parquet
Cloud & Data Engineering	AWS (S3, EC2, Glue, Athena, Bedrock), ETL Pipelines, Apache Spark, Serverless Architecture, FastAPI, Flask, REST APIs
MLOps & DevOps	Docker, Git, CI/CD Deployment Gates, Model Versioning, Redis Caching, Circuit Breakers, Rate Limiting, Regression Testing, Observability & Tracing
Leadership & Methods	Agile/Scrum, Sprint Planning, Cross-functional Team Leadership, Technical Documentation

PROFESSIONAL EXPERIENCE

Data Scientist - ChargeMOD

Trivandrum, Kerala | May 2025 - Present

Customer Support AI Agent

- Architected a modular monolith support platform with a three-path LLM reasoning pipeline: Path A (direct log-based diagnosis), Path B (hybrid BM25 + ChromaDB RAG with cross-encoder reranking), and Path C (dynamic guided questioning), reducing average issue resolution time by approximately 50%.
- Built an entity-cache validation layer that verifies all charger IDs and wallet balances against live API responses before LLM injection, structurally preventing hallucination in a safety-critical environment where the agent can remotely start/stop real chargers.
- Delivered fuzzy charger name resolution (exact, partial, fuzzy, location-based) and real-time SSE streaming for operator dashboards; new support staff handle complex charger faults independently from day one, significantly reducing onboarding time.

Client AI Agent - Natural Language Analytics

- Designed a conversational analytics agent for 6,000+ EV station owners using LLM tool-calling over a registry of 15-20 parameterised PostgreSQL query functions; owner ID injected automatically from authenticated session, making cross-account data access structurally impossible at the DB layer.
- Implemented Redis caching, nightly batch pre-aggregation, circuit breakers for LLM outages, per-owner rate limiting, and prompt regression tests that block deployments when accuracy falls below 95%, serving 1,700+ daily active users.
- Achieved near-zero time-to-insight for energy, session, and fault analytics; eliminated manual dashboard navigation entirely for station owners.

Dynamic Tariff & ML Pricing System

- Built a nightly two-level ML pricing pipeline with network-wide demand forecasting (XGBoost, Random Forest, Facebook Prophet with auto best-model selection) and per-charger pricing using a two-stage model: an activity classifier paired with a log-space energy regressor trained only on active hours, suppressing price volatility for low-usage chargers.
- Applied KMeans behavioural clustering on 36-feature charger profiles (silhouette-optimised k from 3-12), p95 outlier clipping, log1p transformation, and asymmetric sigmoid surge mapping within regulatory caps (+12% surcharge / -6% discount, guaranteed minimum discount hours daily).
- Delivered approximately 10% revenue uplift through demand-aware peak-hour load management across the charging network.

Customer Retention & Churn Reduction

- Built a monthly churn detection system with energy-drop threshold segmentation (10 kWh), rolling-percentile lifecycle clustering across 5 tiers (Low to Inactive) tracking 5-month user trajectories, and a proactive calling console for customer care teams.
- Delivered dual output: a Power BI management dashboard (MAU, retention rate, reactivation trends) and a care-team console with call outcome logging, recovering churned revenue at a fraction of new acquisition cost.

Cloud ETL Pipeline & Power BI Dashboard

- Designed a nightly AWS Glue + S3 Parquet + Athena serverless pipeline extracting EV charging transactions from MongoDB, transforming with PySpark (timestamp correction, duplicate removal, session revenue calculation); date partitioning reduced cloud query costs by approximately 95%.
- Built idempotent partition overwrite logic, a 48-hour lookback buffer for late-arriving data, and strict type casting with a quarantine table for schema anomalies; Power BI auto-refreshes daily, reducing data processing time by 30% while processing 1M+ transactions nightly.

Maintenance & Uptime Monitoring System

- Built a nightly Python batch job computing gap-based uptime/downtime per charger from MongoDB machine logs; gaps over 15 minutes flagged as downtime with idempotent charger-date keying for safe reruns.
- Handled schema divergence between AC and DC charger log collections (extra logger-ID mapping for DC); surfaced in an area-wise drill-down console, contributing to approximately 20% reduction in operational downtime.

Data Science Intern - ChargeMOD

Trivandrum, Kerala | Mar 2024 - May 2024

- Developed an EV Assistant Chatbot using OpenAI function calling, embedding-based retrieval, dynamic entity extraction, and real-time SSE streaming for route optimisation and live charger discovery; the project evolved into the production Customer Support AI Agent.

RPA Data Analyst - Software Incubator Pvt. Ltd.

Kerala | Jun 2022 - May 2024

- Automated document extraction and data processing workflows using JavaScript, Puppeteer, and Kofax; built browser automation pipelines for structured web scraping and data indexing into MySQL databases.

KEY ACHIEVEMENTS

- Reduced customer issue resolution time by approximately 50% through a 3-path AI diagnostic pipeline; new support staff handle complex charger faults independently from day one.
- Enabled self-service natural-language analytics for 6,000+ EV station owners, reaching 1,700+ daily active users and reducing time-to-insight to near-zero.
- Cut cloud query costs by approximately 95% via a Parquet-partitioned AWS ETL pipeline processing 1M+ transactions nightly, with 30% faster data processing.
- Generated approximately 10% revenue uplift through demand-aware ML forecasting and behavioural analytics enabling peak-hour load management.
- Reduced DC charger operational downtime by approximately 20% through automated nightly gap-based downtime detection.

ACADEMIC & RESEARCH PROJECTS

M.Tech Thesis — Lip Reading using Deep Learning

- Designed an audio-free visual speech recognition system using 3D-CNN spatiotemporal feature extraction and BiLSTM sequence decoding trained with CTC loss on the GRID Corpus.
- Achieved a Character Error Rate (CER) of 2.95% and Word Error Rate (WER) of 10.11%, demonstrating accurate lip-reading without any audio input.
- Technologies used: Python, TensorFlow, OpenCV.

Other Academic Projects

- Fruit Picker Robot with computer vision-based object detection and robotic arm sorting.
- Fashion MNIST ANN Classifier achieving 89% accuracy on garment image classification.
- Image-to-Cartoon Generation pipeline using edge detection and colour quantisation techniques.
- 3D Robotic Arm Web Application using Three.js for interactive browser-based simulation and control.

Publications

- “Machine Learning Models for Gamma-Hadron Separation,” CREEST International Conference — Gradient Boosting ensemble for gamma-ray vs. hadronic background classification that outperformed SVM, AdaBoost, Naive Bayes, and Decision Tree baselines.

EDUCATION

M.Tech in Computer Science (Digital Image Computing)

2023 - 2025

Kerala University, Karyavattom

B.Tech in Computer Science and Engineering

2015 - 2019

University College of Engineering, Karyavattom

CERTIFICATIONS & ACTIVITIES

- The Complete Python Bootcamp: From Zero to Hero - Udemy
- Data Visualization with Power BI - Great Learning
- Co-led an AI, Machine Learning & Robotics Bootcamp at Tella Academy for high school students